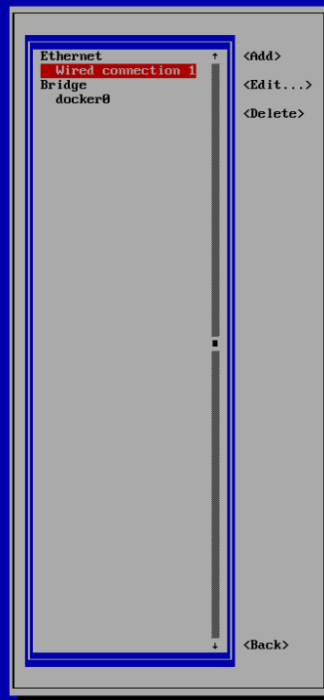
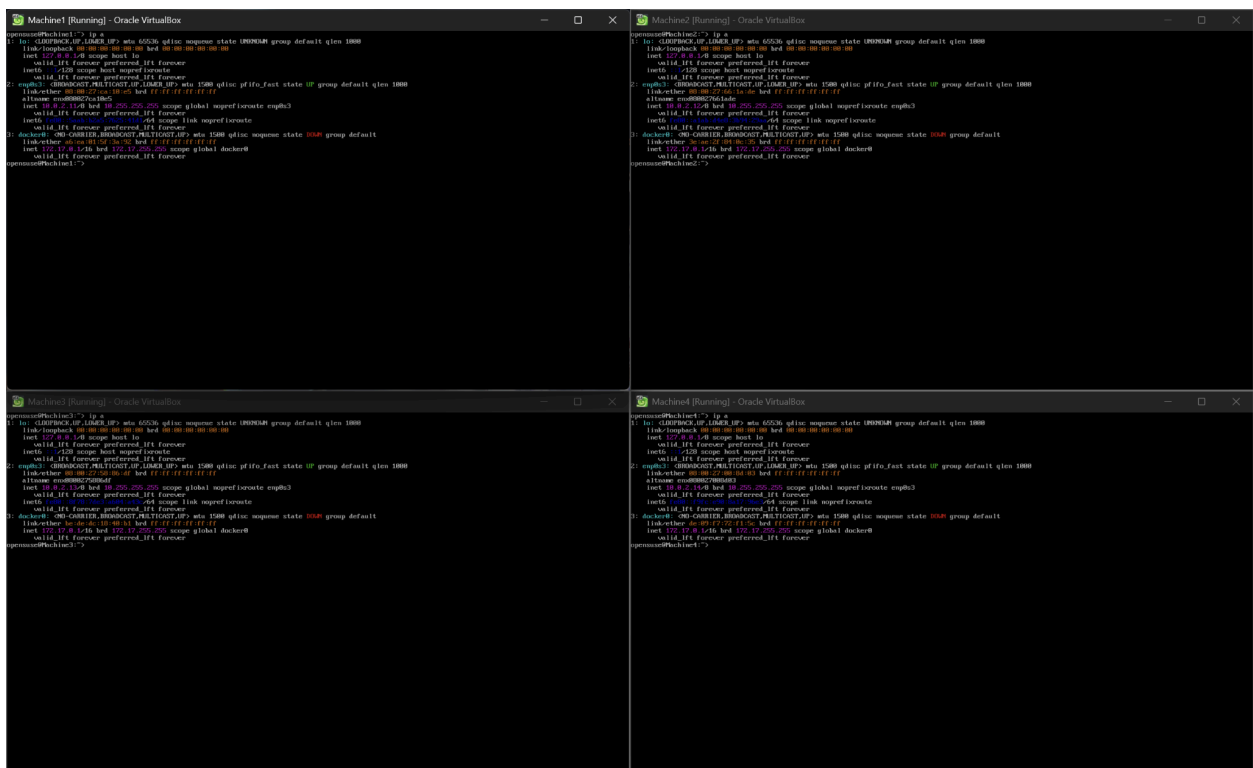
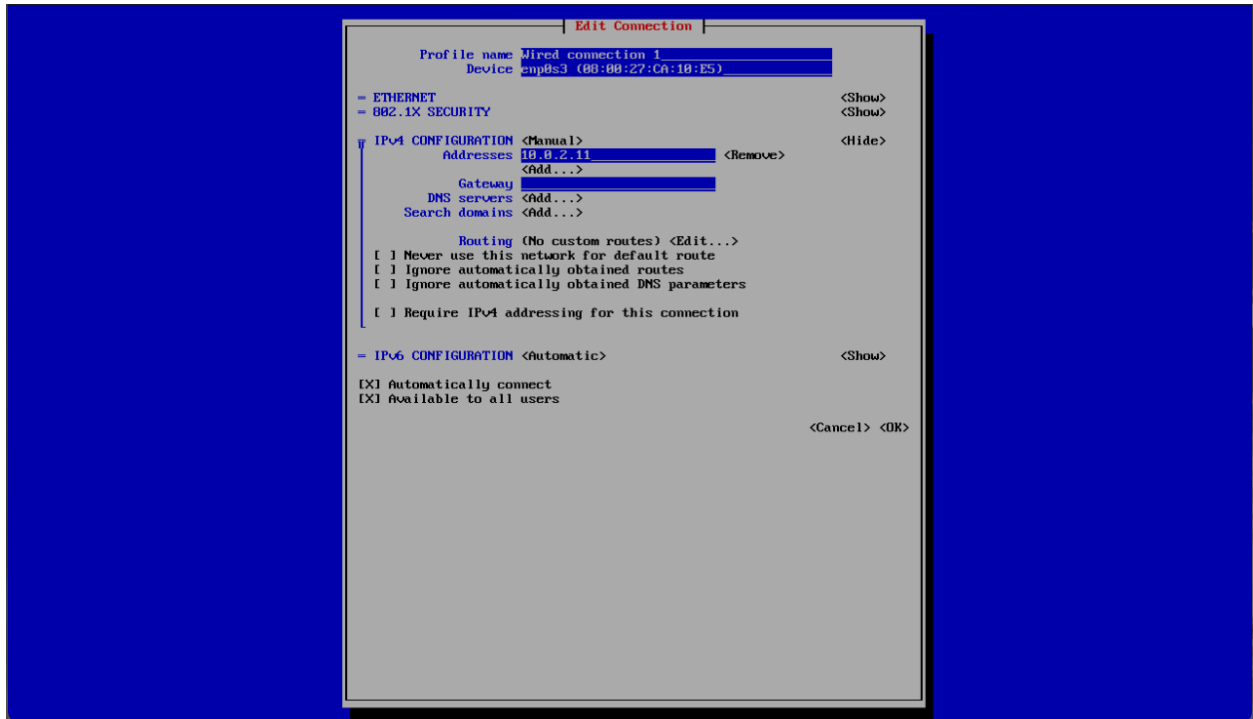






```
GNU nano 8.6 /etc/hosts
# hosts      This file describes a number of hostname-to-address
#            mappings for the TCP/IP subsystem.  It is mostly
#            used at boot time, when no name servers are running.
#            On small systems, this file can be used instead of a
#            "named" name server.
# Syntax:
#
# IP-Address Full-Qualified-Hostname Short-Hostname
#
127.0.0.1    localhost localhost.localdomain
::1         localhost localhost.localdomain ipv6-localhost ipv6-loopback
# special IPv6 addresses
fe80::      ipv6-localhost
ff00::      ipv6-mcastprefix
ff02::1     ipv6-allnodes
ff02::2     ipv6-allrouters
ff02::3     ipv6-allhosts
10.0.2.12   Machine2
10.0.2.13   Machine3
10.0.2.14   Machine4

[ Read 25 lines ]
Help      Write Out  Where Is  Cut       Execute   Location  Undo      Set Mark  To Bracket Previous
Exit      Read File   Replace  Paste     Justify   Go To Line Redo      Copy      Where Was  Next

opensuse@Machine1:~$ ping Machine2
PING Machine2 (10.0.2.12): 56(84) bytes of data.
64 bytes from Machine2 (10.0.2.12): icmp_seq=1 ttl=64 time=5.79 ms
64 bytes from Machine2 (10.0.2.12): icmp_seq=2 ttl=64 time=2.45 ms
64 bytes from Machine2 (10.0.2.12): icmp_seq=3 ttl=64 time=2.38 ms
64 bytes from Machine2 (10.0.2.12): icmp_seq=4 ttl=64 time=138 ms
^C
-- Machine2 ping statistics --
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 2.383/37.282/138.193/59.323 ms
opensuse@Machine1:~$ ping Machine3
PING Machine3 (10.0.2.13): 56(84) bytes of data.
64 bytes from Machine3 (10.0.2.13): icmp_seq=1 ttl=64 time=6.22 ms
64 bytes from Machine3 (10.0.2.13): icmp_seq=2 ttl=64 time=2.57 ms
64 bytes from Machine3 (10.0.2.13): icmp_seq=3 ttl=64 time=2.69 ms
64 bytes from Machine3 (10.0.2.13): icmp_seq=4 ttl=64 time=2.58 ms
^C
-- Machine3 ping statistics --
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 2.497/3.493/6.219/1.575 ms
opensuse@Machine1:~$ ping Machine4
PING Machine4 (10.0.2.14): 56(84) bytes of data.
64 bytes from Machine4 (10.0.2.14): icmp_seq=1 ttl=64 time=5.88 ms
64 bytes from Machine4 (10.0.2.14): icmp_seq=2 ttl=64 time=3.88 ms
64 bytes from Machine4 (10.0.2.14): icmp_seq=3 ttl=64 time=2.59 ms
64 bytes from Machine4 (10.0.2.14): icmp_seq=4 ttl=64 time=2.46 ms
^C
-- Machine4 ping statistics --
4 packets transmitted, 4 received, 0% packet loss, time 3009ms
rtt min/avg/max/mdev = 2.457/3.479/4.999/1.838 ms
opensuse@Machine1:~$ _
```


1. Config Servers (all 3 VMs)
↓
2. rs.initiate() the Config Replica Set
↓
3. Shard Servers (all 3 VMs)
↓
4. rs.initiate() the Shard Replica Set
↓
5. Mongos Router
↓
6. sh.addShard()

```
config-machine2-data
Machine2:/home/opensuse # docker run -d --name config-Machine2 -p 27019:27019 -v config-Machine2-data:/data/db mongodb/mongodb-enterprise-server:latest \
> mongod --configsvr --replSet configRS --port 27019
Unable to find image 'mongodb/mongodb-enterprise-server:latest' locally
docker: Error response from daemon: Get "https://registry-1.docker.io/v2/": context deadline exceeded
Run 'docker run --help' for more information
Machine2:/home/opensuse #
```

```
opensuse@Machine3:~> su
Password:
Machine3:/home/opensuse # docker volume create config-Machine3-data
config-Machine3-data
Machine3:/home/opensuse #
```

```
Using default tag: latest
latest: Pulling from library/mongo
b40150c1c271: Pull complete
6126dc5fffed: Pull complete
ccce04f560d7: Pull complete
f39dec9b5448: Pull complete
0b1042dab2ed: Pull complete
ba27587cdb34: Pull complete
51c17794a12f: Pull complete
22dc34102e26: Pull complete
Digest: sha256:7abfba0407c9330373f8173981ea4d09cd8a82cdf0e06ccaf7008048d1d24f62
Status: Downloaded newer image for mongo:latest
docker.io/library/mongo:latest
Machine2:/home/opensuse # _
```

```
Machine2:/home/opensuse # docker run -d --name config-Machine2 -p 27019:27019 -v config-Machine2-data:/data/db mongo mongod --config --replSet configRS \
> --port 27019
b9abd43b5501c00174ae4d2bfe5339218dc4208b0a796ed6d5cedf523087027d
Machine2:/home/opensuse #
Machine2:/home/opensuse # _
```

```
Machine2:/home/opensuse # docker volume create shard-Machine2-data
shard-Machine2-data
Machine2:/home/opensuse # docker run -d --name shard-Machine2 -p 27018:27018 -v shard-Machine2-data:/data/db mongo mongod --shardsvr \
> --replSet --port 27018
92b36371d0ef483d14ae8d4b03ae874ac749ab76ae97d511f4a65886f669237e
Machine2:/home/opensuse #
```

File Edit Selection View Go Run Terminal Help

← → | GitLab2

names.txt #4 Add my name to the... X

Add my name to the list #4

Open LNewell00 wants to merge changes into bbright32/GitLab5:main from bbright32/GitLab5:LoganNewell-Add

Checkout 'main' Refresh

LNewell00 commented seconds ago

Add my name via my new branch

♦ Add my name to the list 9711afb 7 minutes ago

✓ This branch has no conflicts with the base branch.

Leave a comment

Close Pull Request Comment

REVIEW PULL REQUEST #4

Leave a comment

Comment

This branch has no conflicts with the base branch.

Checkout 'main'

Reviewers

None yet

Still in progress? Convert to draft

Assignees

None yet

Labels

None yet

Project

Sign in with more permissions to see projects

Milestone

No milestone

NOTIFICATIONS

Would you like Visual Studio Code to periodically run "git fetch"?

Source: Git

Yes No Ask Me Later

LoganNewell-Add 0 Pull Request #4

```

Machine2:/home/opensuse # docker exec -it config-Machine2 mongo --port 27019
MongoDB shell version v4.4.30
connecting to: mongodb://127.0.0.1:27019/?compressors=disabled&gssapiServiceName=mongod
b
Implicit session: session { "id" : UUID("dc19a656-6ab2-49b9-acc3-844711e54202") }
MongoDB server version: 4.4.30
Welcome to the MongoDB shell.
For interactive help, type "help".
For more comprehensive documentation, see
https://docs.mongodb.com/
Questions? Try the MongoDB Developer Community Forums
https://community.mongodb.com
---
The server generated these startup warnings when booting:
  2026-05-01T17:15:51.743+00:00: Access control is not enabled for the database.
Read and write access to data and configuration is unrestricted
  2026-05-01T17:15:51.747+00:00: /sys/kernel/mm/transparent_hugepage/enabled is '
always'. We suggest setting it to 'never'
---
> rs.initiate({
...   _id: "configRS",
...   configsvr: true,
...   members: [
...     { _id: 0, host: "10.0.2.12:27019" },
...     { _id: 1, host: "10.0.2.13:27019" },
...     { _id: 2, host: "10.0.2.14:27019" }
...   ]
... })
{
  "ok" : 1,
  "$gleStats" : {
    "lastOpTime" : Timestamp(1777656395, 1),
    "electionId" : ObjectId("0000000000000000000000000000")
  },
  "lastCommittedOpTime" : Timestamp(0, 0)
}
configRS:SECONDARY> |

```



```

Machine2:/home/opensuse # docker exec -it shard-Machine2 mongo --port 27018
MongoDB shell version v4.4.30
connecting to: mongodb://127.0.0.1:27018/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("cac30f37-822e-4837-90ca-4ceee7ac2a7a") }
MongoDB server version: 4.4.30
Welcome to the MongoDB shell.
For interactive help, type "help".
For more comprehensive documentation, see
  https://docs.mongodb.com/
Questions? Try the MongoDB Developer Community Forums
  https://community.mongodb.com
---
The server generated these startup warnings when booting:
  2026-05-01T17:15:52.039+00:00: Access control is not enabled for the database.
Read and write access to data and configuration is unrestricted
  2026-05-01T17:15:52.043+00:00: /sys/kernel/mm/transparent_hugepage/enabled is '
always'. We suggest setting it to 'never'
---
> rs.initiate({
...   _id: "shardRS",
...   members: [
...     { _id: 0, host: "10.0.2.12:27018" },
...     { _id: 1, host: "10.0.2.13:27018" },
...     { _id: 2, host: "10.0.2.14:27018" }
...   ]
... })
{ "ok" : 1 }
shardRS:SECONDARY> |

```

```

Machine1:/home/opensuse # docker run -d --restart unless-stopped --name mongos-Machine1 -p 27017:27017 \
> mongo:4.4 \
> mongos --configdb configRS/10.0.2.12:27019,10.0.2.13:27019,10.0.2.14:27019 \
> --bind_ip_all --port 27017
e2a24cab4a4973bc0d62007d2dbdc50dab6af72b38ba709bc415b89ba93c76ae
Machine1:/home/opensuse # docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
e2a24cab4a49   mongo:4.4 "docker-entrypoint.s..." 4 seconds ago  Up 3 seconds  0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp  mongos-Machine1
Machine1:/home/opensuse # |

```

```
Machine1:/home/opensuse # docker exec -it mongos-Machine1 mongo --port 27017
MongoDB shell version v4.4.30
connecting to: mongodb://127.0.0.1:27017/?compressors=disabled&gssapiServiceName=mongod
b
Implicit session: session { "id" : UUID("abe4a090-405c-4e56-ba79-1c681b15688a") }
MongoDB server version: 4.4.30
Welcome to the MongoDB shell.
For interactive help, type "help".
For more comprehensive documentation, see
  https://docs.mongodb.com/
Questions? Try the MongoDB Developer Community Forums
  https://community.mongodb.com
---
The server generated these startup warnings when booting:
  2026-05-01T17:27:54.752+00:00: Access control is not enabled for the database.
Read and write access to data and configuration is unrestricted
---
mongos> sh.addShard("shardRS/10.0.2.12:27018,10.0.2.13:27018,10.0.2.14:27018")
{
  "shardAdded" : "shardRS",
  "ok" : 1,
  "operationTime" : Timestamp(1777656501, 3),
  "$clusterTime" : {
    "clusterTime" : Timestamp(1777656501, 3),
    "signature" : {
      "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAAA="),
      "keyId" : NumberLong(0)
    }
  }
}
mongos> |
```

```

Machine1:/home/opensuse # docker exec -it mongos-Machine1 mongo --port 27017
MongoDB shell version v4.4.30
connecting to: mongod://127.0.0.1:27017/?compressors=disabled&gssapiServiceName=mongod
b
Implicit session: session { "id" : UUID("3b598319-4d4b-4dec-bb8d-ba229916c8d8") }
MongoDB server version: 4.4.30
---
The server generated these startup warnings when booting:
    2026-05-01T17:27:54.752+00:00: Access control is not enabled for the database.
Read and write access to data and configuration is unrestricted
---
mongos> sh.status()
--- Sharding Status ---
  sharding version: {
    "_id" : 1,
    "minCompatibleVersion" : 5,
    "currentVersion" : 6,
    "clusterId" : ObjectId("69f4e25788d776692e53e97b")
  }
  shards:
    { "_id" : "shardRS", "host" : "shardRS/10.0.2.12:27018,10.0.2.13:27018,10.0.2.14:27018", "state" : 1 }
  active mongoses:
    "4.4.30" : 1
  autosplit:
    Currently enabled: yes
  balancer:
    Currently enabled:  yes
    Currently running:  no
    Failed balancer rounds in last 5 attempts:  0
    Migration Results for the last 24 hours:
      No recent migrations
  databases:
    { "_id" : "config", "primary" : "config", "partitioned" : true }
      config.system.sessions
        shard key: { "_id" : 1 }
        unique: false
        balancing: true
        chunks:
          shardRS 1024
      too many chunks to print, use verbose if you want to force print
t
mongos> |

```

```

Machine2:/home/opensuse # docker exec -it shard-Machine2 mongo --port 27018
MongoDB shell version v4.4.30
connecting to: mongodb://127.0.0.1:27018/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("3be0ee26-a7f5-4bae-bdbe-ced6a014d34b") }
MongoDB server version: 4.4.30
---
The server generated these startup warnings when booting:
  2026-05-01T17:15:52.039+00:00: Access control is not enabled for the database. Read and write access to
data and configuration is unrestricted
  2026-05-01T17:15:52.043+00:00: /sys/kernel/mm/transparent_hugepage/enabled is 'always'. We suggest sett
ing it to 'never'
---
shardRS:PRIMARY> rs.status()
{
  "set" : "shardRS",
  "date" : ISODate("2026-05-01T17:30:59.395Z"),
  "myState" : 1,
  "term" : NumberLong(1),
  "syncSourceHost" : "",
  "syncSourceId" : -1,
  "heartbeatIntervalMillis" : NumberLong(2000),
  "majorityVoteCount" : 2,
  "writeMajorityCount" : 2,
  "votingMembersCount" : 3,
  "writableVotingMembersCount" : 3,
  "optimes" : {
    "lastCommittedOpTime" : {
      "ts" : Timestamp(1777656656, 1),
      "t" : NumberLong(1)
    },
    "lastCommittedWallTime" : ISODate("2026-05-01T17:30:56.202Z"),
    "readConcernMajorityOpTime" : {
      "ts" : Timestamp(1777656656, 1),
      "t" : NumberLong(1)
    },
    "readConcernMajorityWallTime" : ISODate("2026-05-01T17:30:56.202Z"),
    "appliedOpTime" : {
      "ts" : Timestamp(1777656656, 1),
      "t" : NumberLong(1)
    },
    "durableOpTime" : {
      "ts" : Timestamp(1777656656, 1),
      "t" : NumberLong(1)
    },
    "lastAppliedWallTime" : ISODate("2026-05-01T17:30:56.202Z"),
    "lastDurableWallTime" : ISODate("2026-05-01T17:30:56.202Z")
  },
  "lastStableRecoveryTimestamp" : Timestamp(1777656636, 1),
  "electionCandidateMetrics" : {
    "lastElectionReason" : "electionTimeout",
    "lastElectionDate" : ISODate("2026-05-01T17:27:35.881Z"),
    "electionTerm" : NumberLong(1),
    "lastCommittedOpTimeAtElection" : {
      "ts" : Timestamp(0, 0),
      "t" : NumberLong(-1)
    },
    "lastSeenOpTimeAtElection" : {
      "ts" : Timestamp(1777656444, 1),
      "t" : NumberLong(-1)
    },
    "numVotesNeeded" : 2,
    "priorityAtElection" : 1,
    "electionTimeoutMillis" : NumberLong(10000),
    "numCatchUpOps" : NumberLong(0),
    "newTermStartDate" : ISODate("2026-05-01T17:27:36.065Z"),
    "wMajorityWriteAvailabilityDate" : ISODate("2026-05-01T17:27:36.734Z")
  },
  "members" : [
    {
      "_id" : 0,
      "name" : "10.0.2.12:27018",
      "health" : 1,

```

```

        "t" : NumberLong(1)
    },
    "optimeDurable" : {
        "ts" : Timestamp(1777656656, 1),
        "t" : NumberLong(1)
    },
    "optimeDate" : ISODate("2026-05-01T17:30:56Z"),
    "optimeDurableDate" : ISODate("2026-05-01T17:30:56Z"),
    "lastAppliedWallTime" : ISODate("2026-05-01T17:30:56.202Z"),
    "lastDurableWallTime" : ISODate("2026-05-01T17:30:56.202Z"),
    "lastHeartbeat" : ISODate("2026-05-01T17:30:57.940Z"),
    "lastHeartbeatRecv" : ISODate("2026-05-01T17:30:58.042Z"),
    "pingMs" : NumberLong(2),
    "lastHeartbeatMessage" : "",
    "syncSourceHost" : "10.0.2.12:27018",
    "syncSourceId" : 0,
    "infoMessage" : "",
    "configVersion" : 2,
    "configTerm" : 1
},
{
    "_id" : 2,
    "name" : "10.0.2.14:27018",
    "health" : 1,
    "state" : 2,
    "stateStr" : "SECONDARY",
    "uptime" : 214,
    "optime" : {
        "ts" : Timestamp(1777656656, 1),
        "t" : NumberLong(1)
    },
    "optimeDurable" : {
        "ts" : Timestamp(1777656656, 1),
        "t" : NumberLong(1)
    },
    "optimeDate" : ISODate("2026-05-01T17:30:56Z"),
    "optimeDurableDate" : ISODate("2026-05-01T17:30:56Z"),
    "lastAppliedWallTime" : ISODate("2026-05-01T17:30:56.202Z"),
    "lastDurableWallTime" : ISODate("2026-05-01T17:30:56.202Z"),
    "lastHeartbeat" : ISODate("2026-05-01T17:30:57.943Z"),
    "lastHeartbeatRecv" : ISODate("2026-05-01T17:30:57.966Z"),
    "pingMs" : NumberLong(2),
    "lastHeartbeatMessage" : "",
    "syncSourceHost" : "10.0.2.12:27018",
    "syncSourceId" : 0,
    "infoMessage" : "",
    "configVersion" : 2,
    "configTerm" : 1
}
],
"ok" : 1,
"$gleStats" : {
    "lastOpTime" : Timestamp(0, 0),
    "electionId" : ObjectId("7fffffff000000000000000001")
},
"lastCommittedOpTime" : Timestamp(1777656656, 1),
"$configServerState" : {
    "opTime" : {
        "ts" : Timestamp(1777656650, 1),
        "t" : NumberLong(1)
    }
},
"$clusterTime" : {
    "clusterTime" : Timestamp(1777656656, 1),
    "signature" : {
        "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAAA="),
        "keyId" : NumberLong(0)
    }
},
"operationTime" : Timestamp(1777656656, 1)
}
shardRS:PRIMARY> |

```

```
mongos> // Enable sharding on a test database
mongos> sh.enableSharding("testdb")
{
  "ok" : 1,
  "operationTime" : Timestamp(1777656778, 3),
  "$clusterTime" : {
    "clusterTime" : Timestamp(1777656778, 3),
    "signature" : {
      "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAAA="),
      "keyId" : NumberLong(0)
    }
  }
}
mongos> |
```

```
mongos> // Create a sharded collection
mongos> sh.shardCollection("testdb.testcol", { _id: "hashed" })
{
  "collectionsharded" : "testdb.testcol",
  "collectionUUID" : UUID("d78117f2-9488-425e-851a-e3b33ef1cea2"),
  "ok" : 1,
  "operationTime" : Timestamp(1777656796, 9),
  "$clusterTime" : {
    "clusterTime" : Timestamp(1777656796, 9),
    "signature" : {
      "hash" : BinData(0,"AAAAAAAAAAAAAAAAAAAAAAAAAAAA="),
      "keyId" : NumberLong(0)
    }
  }
}
mongos> |
```

```
mongos> // Insert some data
mongos> use testdb
switched to db testdb
mongos> for (let i = 0; i < 100; i++) {
... db.testcol.insertOne({ name: "test" + i, value: i })
... }
{
  "acknowledged" : true,
  "insertedId" : ObjectId("69f4e4240eae0f5ec0235f5e")
}
mongos> db.testcol.getShardDistribution()

Shard shardRS at shardRS/10.0.2.12:27018,10.0.2.13:27018,10.0.2.14:27018
data : 5KiB docs : 100 chunks : 2
estimated data per chunk : 2KiB
estimated docs per chunk : 50

Totals
data : 5KiB docs : 100 chunks : 2
Shard shardRS contains 100% data, 100% docs in cluster, avg obj size on shard : 53B
mongos> |
```

```
shardRS:PRIMARY> use testdb
switched to db testdb
shardRS:PRIMARY> db.testcol.find().pretty()
{
  "_id" : ObjectId("69f4e4230eae0f5ec0235efb"),
  "name" : "test0",
  "value" : 0
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235efc"),
  "name" : "test1",
  "value" : 1
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235efd"),
  "name" : "test2",
  "value" : 2
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235efe"),
  "name" : "test3",
  "value" : 3
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235eff"),
  "name" : "test4",
  "value" : 4
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235f00"),
  "name" : "test5",
  "value" : 5
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235f01"),
  "name" : "test6",
  "value" : 6
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235f02"),
  "name" : "test7",
  "value" : 7
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235f03"),
  "name" : "test8",
  "value" : 8
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235f04"),
  "name" : "test9",
  "value" : 9
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235f05"),
  "name" : "test10",
  "value" : 10
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235f06"),
  "name" : "test11",
  "value" : 11
}

{
  "_id" : ObjectId("69f4e4230eae0f5ec0235f07"),
  "name" : "test12",
  "value" : 12
}
```